

Udayan Atreya

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EDUCATION

Master of Science, Computer Science

Expected May 2026

San Jose State University, San Jose, US

Coursework: Retrieval Augmented Generation, LLM Prompt Engineering & Fine-Tuning, Applied Deep Learning.

Bachelor of Technology, Computer Science & Engineering

May 2020

Manipal University, Jaipur, India

Coursework: Data Structures & Algorithms, Relational Database Management Systems, Object-Oriented Programming.

TECHNICAL SKILLS

Frameworks: LangChain, RAG, HuggingFace, PEFT, TensorFlow, PyTorch, FastAPI, RAGAS, Git

LLMs & AI Tools: Claude, GPT-4o, Gemini, Llama-3, Mistral-7B, Ollama, DeepSeek, Groq, Vapi

Cloud & Database: AWS (Lambda, S3, API Gateway), CI/CD, MongoDB, DynamoDB, Pinecone, ChromaDB

Languages: Python, Java, C#, SQL

PROJECT EXPERIENCE

Evaluator-Optimizer RAG (*LangChain, GPT-4o, ChromaDB, RAGAS, Whisper*)

[\[GitHub\]](#)

Aug 2025 – Mar 2026

- Architected a **multi-step agentic RAG framework** with Evaluator-Optimizer feedback loop — designed **prompt templates**, **system instructions**, and **guardrails** for hallucination mitigation, reducing hallucination by **25%** and improving answer accuracy by **18%** over baseline.
- Conducted hands-on ablation study on **embeddings**: benchmarked MiniLM vs. BGE-base embedding models stored in ChromaDB, demonstrating **98% performance parity** at **50% lower storage cost**; tuned chunking strategies and embedding model selection for optimal retrieval quality.
- Built an **automated LLM evaluation and regression test suite** using RAGAS and Llama-3-70b (LLM-as-a-judge) over 500+ STEM QA pairs, quantifying Faithfulness, Context Precision, Context Recall, Noise Sensitivity, and Semantic Similarity.
- Implemented **document retrieval strategies** — Dynamic Chunking, MMR re-ranking, and hybrid semantic search — reducing context redundancy by **40%** while maintaining **90%+ semantic coverage**; optimized token usage and inference latency end-to-end.

Agentic BMC Observability (*LangChain, Gemini 1.5, FastAPI, MongoDB, AWS S3*)

[\[GitHub\]](#)

Jun 2025 – Jul 2025

- Designed a **multi-agent, multi-step workflow** with a semantic router orchestrating specialized sub-agents for query classification, telemetry analysis, and root cause diagnosis — achieving **95% query classification accuracy** and reducing manual triage time by **30%**.
- Implemented **document retrieval strategies** via a hybrid engine (MongoDB hot store + AWS S3 archival) with intelligent routing and ranking, cutting retrieval latency by **75%** (2 min → 30 s) for enterprise-scale operational data.
- Engineered **system instructions** and strict **policy guardrails** with role-based access controls for agentic hardware interventions on Baseboard Management Controllers; enforced **human-in-the-loop validation** preventing **100% of unauthorized state changes**.

Mistral-7B QLoRA Benchmark (*PyTorch, HuggingFace, PEFT, bitsandbytes, TRL*)

[\[GitHub\]](#)

2025

- Engineered structured **prompt templates** and a label-extraction parsing pipeline to elicit discrete NLI class predictions from Mistral-7B; fine-tuned with QLoRA (rank=16, <0.2% of 7.2B parameters), achieving **90.2% accuracy / F1=0.90** on GLUE MNLI — surpassing a RoBERTa-large-mnli discriminative baseline.
- Applied 4-bit NF4 quantization (bitsandbytes) and hands-on **embedding layer** optimization, reducing GPU memory from ~28 GB to ~3.5 GB (**87% reduction**), enabling full fine-tuning on a single A100; benchmarked QLoRA and DoRA adapter variants with systematic hyperparameter sweeps.
- Developed automated **LLM regression test scripts** (`evaluate_roBERTa.py`, `evaluate_baseline_llm.py`, `evaluate_finetuned`) for systematic benchmarking of four model variants on Entailment, Neutral, and Contradiction classes, producing reproducible timestamped evaluation reports.

WORK EXPERIENCE

Software Engineer, Accenture, Bengaluru, India

Dec 2021 – Jul 2024

Associate Software Engineer, Accenture, Bengaluru, India

Nov 2020 – Nov 2021

- Delivered a **ChatOps platform** routing natural language queries through **multi-step automation workflows** via serverless microservices (AWS Lambda, API Gateway), integrating tools and internal APIs to reduce operational overhead by **40%** (~5 min to ~1 min).
- Architected a user notification system with microservices on **AWS (API Gateway, Lambda, DynamoDB)** delivering remediation stage notifications, resulting in a **20% increase in DAU** post-launch.
- Optimized cloud storage by redesigning **DynamoDB** access patterns (schema, pk, sk) and CRUD microservices, **reducing retrieval latency by 37%** for resilient, high-performing workloads.
- Delivered a **code vulnerability remediation** solution using serverless architecture (AWS Lambda, S3, Event-Bridge) and Azure DevOps CI/CD, safeguarding **400+ applications**; designed single-touch server patching resolving ServiceNow alerts and **reducing MTTR by 80%**.